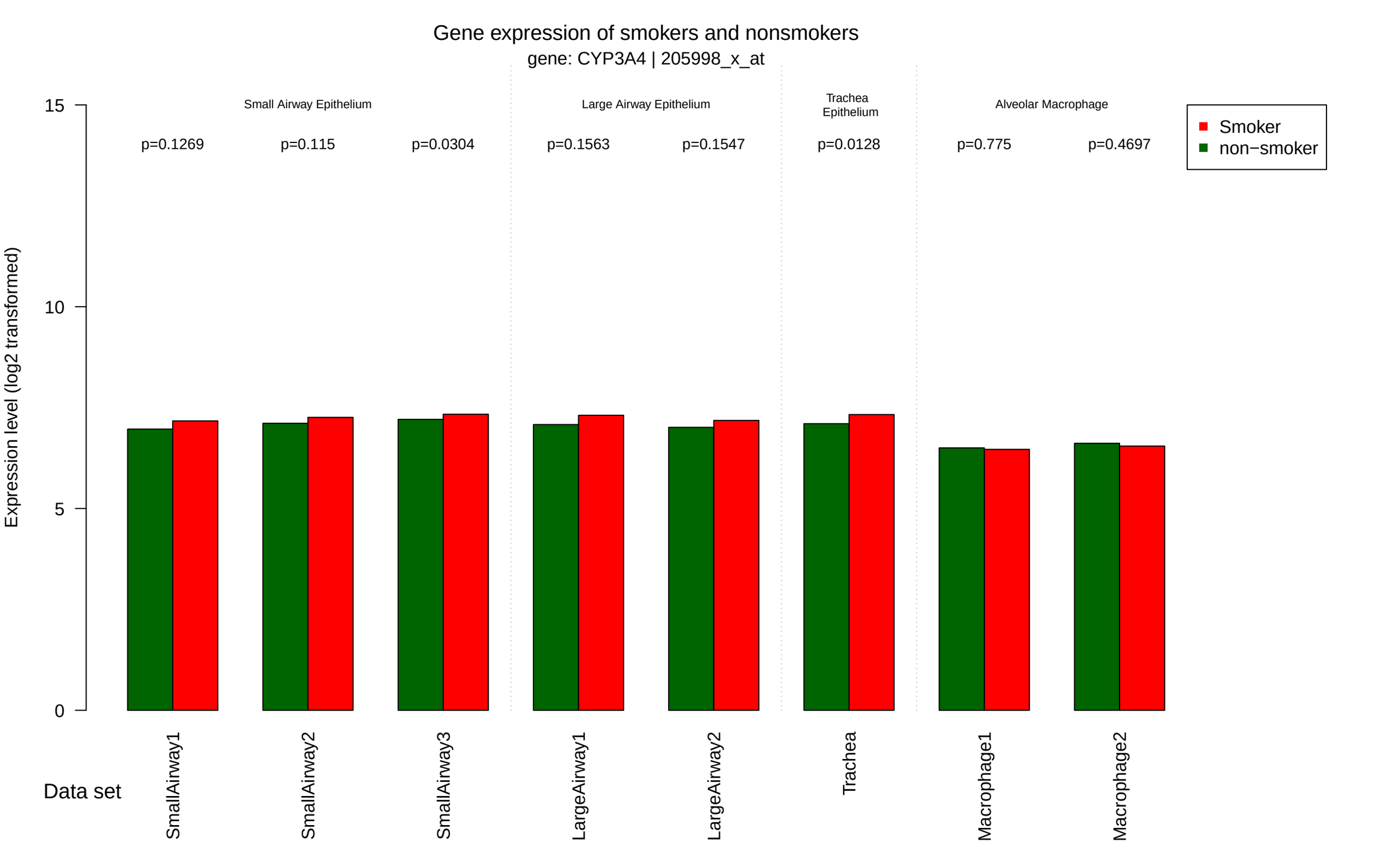
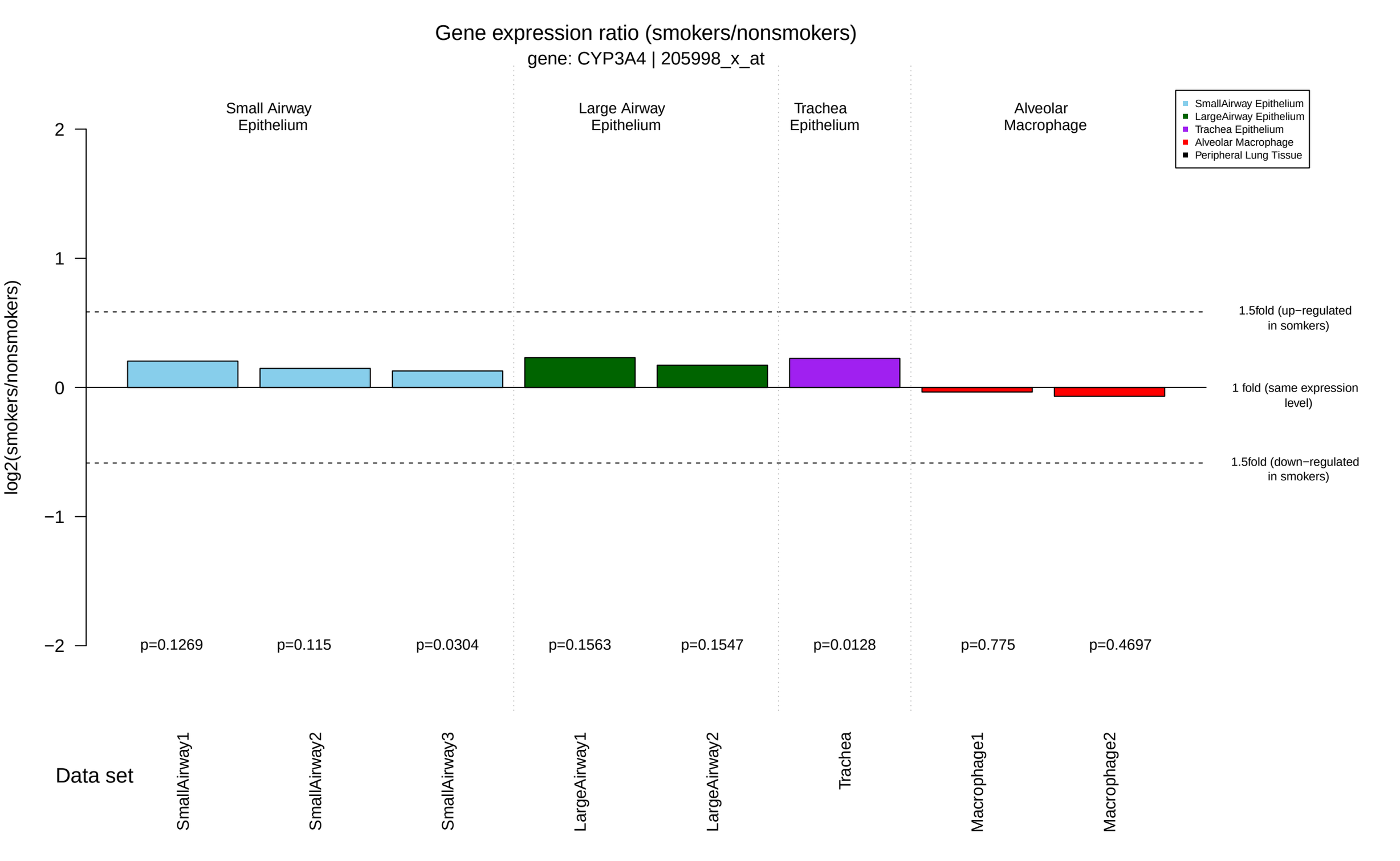
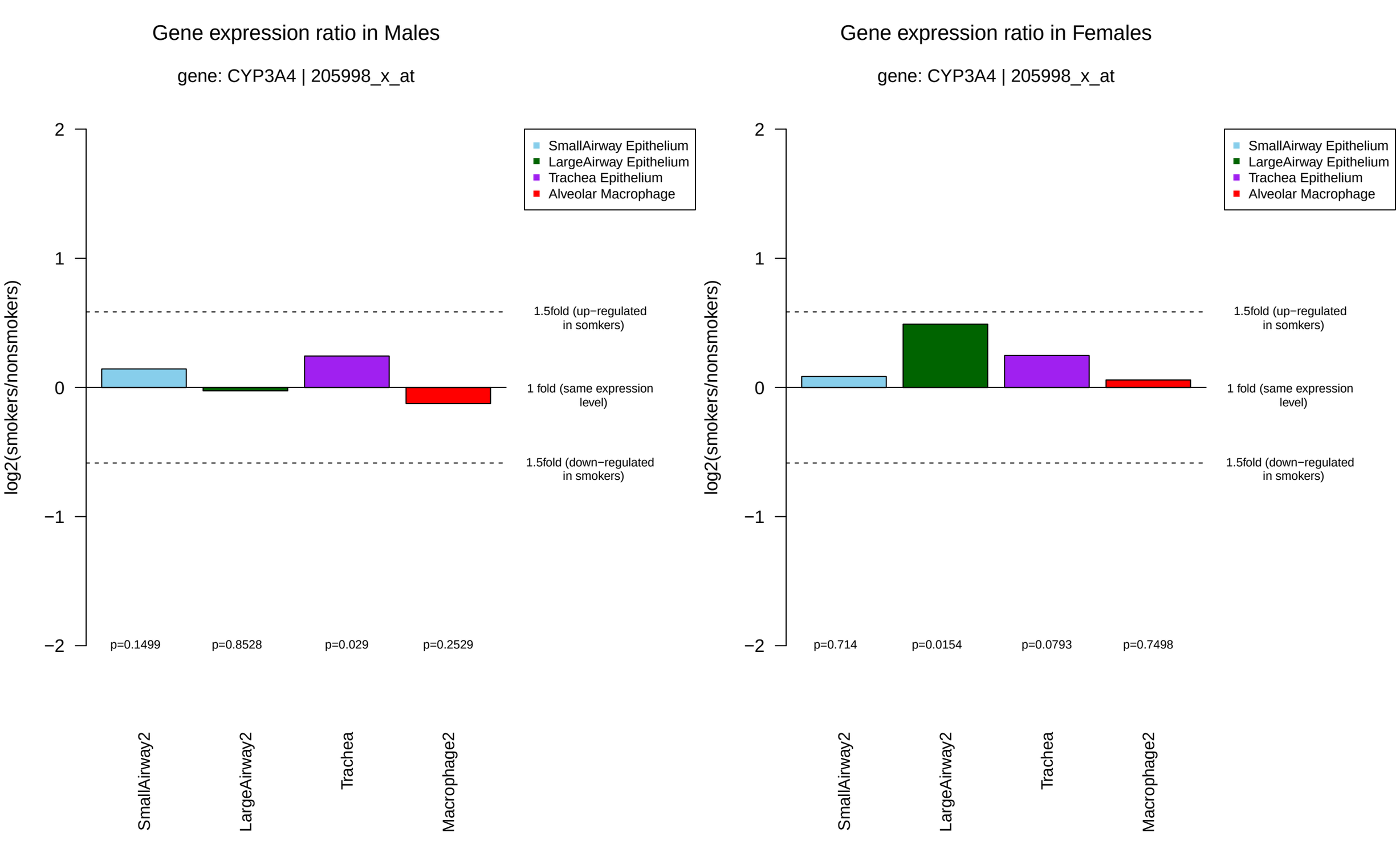
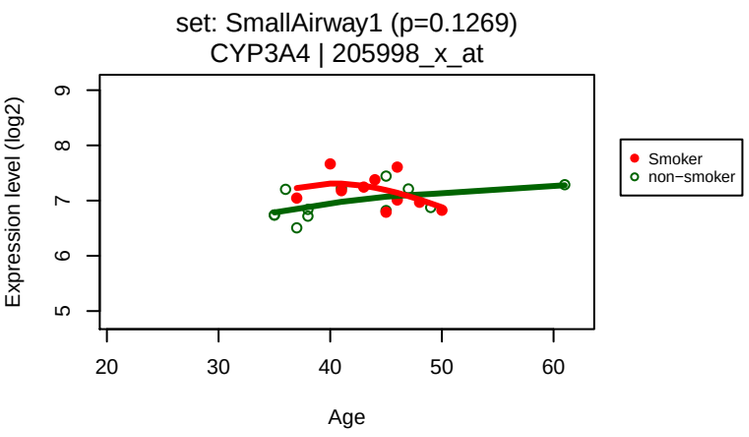
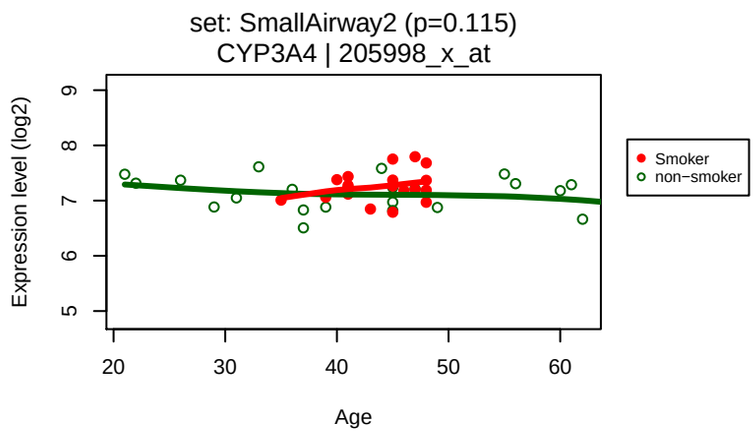
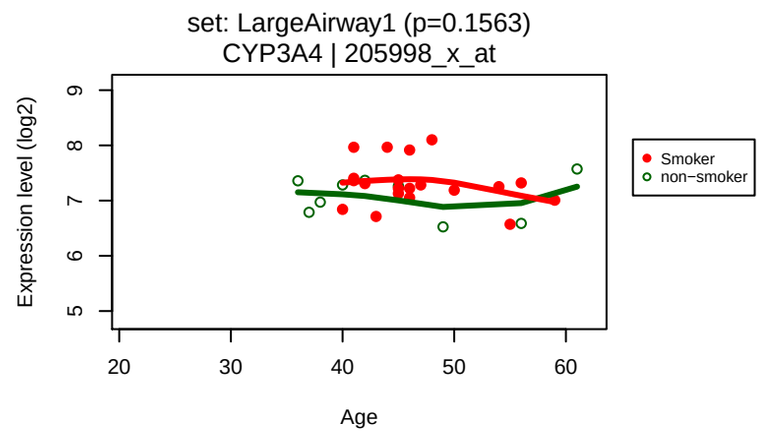

Gene expression of smokers and nonsmokers
gene: CYP3A4 | 205998_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1269
p=0.115
p=0.0304
p=0.1563
p=0.1547
p=0.0128
p=0.775
p=0.4697
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

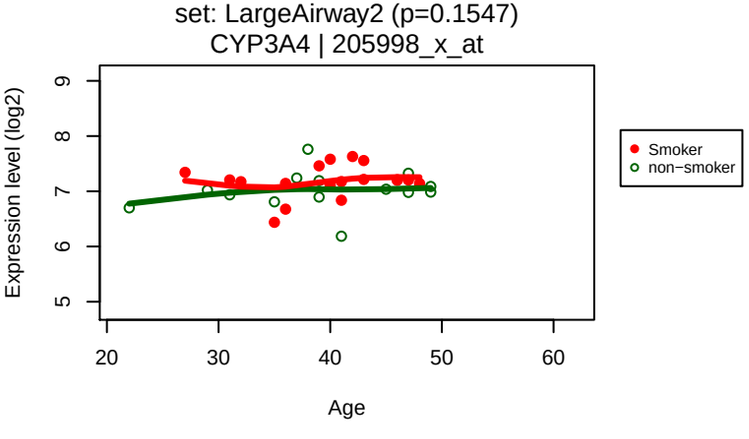

Gene expression ratio (smokers/nonsmokers)
gene: CYP3A4 | 205998_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1269
p=0.115
p=0.0304
p=0.1563
p=0.1547
p=0.0128
p=0.775
p=0.4697
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

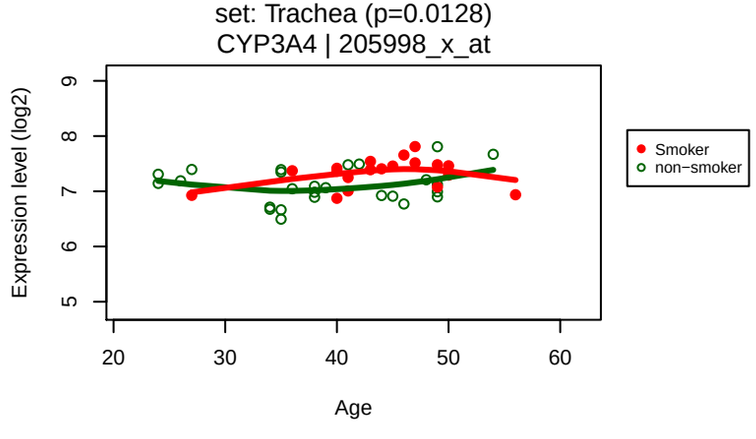

Gene expression ratio in Males
gene: CYP3A4 | 205998_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1499
p=0.8528
p=0.029
p=0.2529
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP3A4 | 205998_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.714
p=0.0154
p=0.0793
p=0.7498
SmallAirway2
LargeAirway2
Trachea
Macrophage2

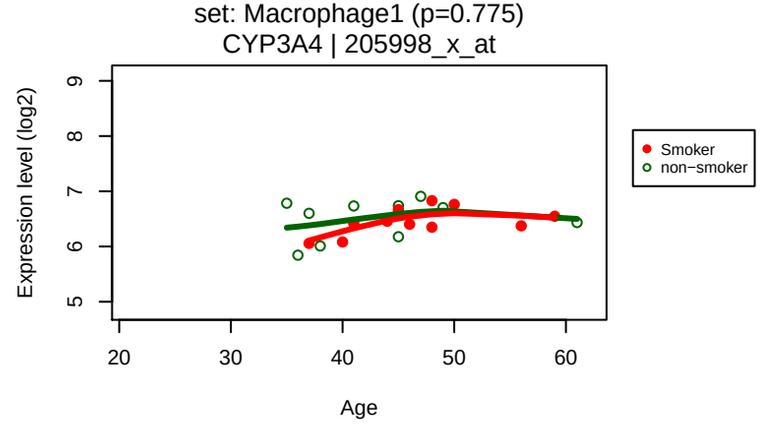

set: SmallAirway1 (p=0.1269)
CYP3A4 | 205998_x_at
Expression level (log2)
Age
Smoker
non−smoker

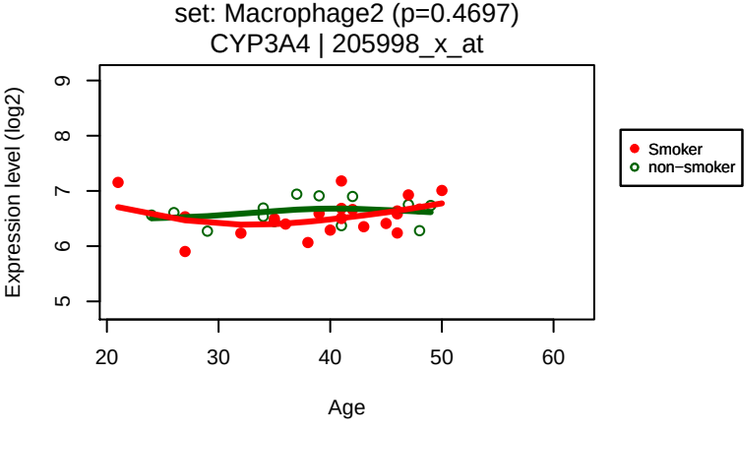

set: SmallAirway2 (p=0.115)
CYP3A4 | 205998_x_at
Expression level (log2)
Age
Smoker
non−smoker

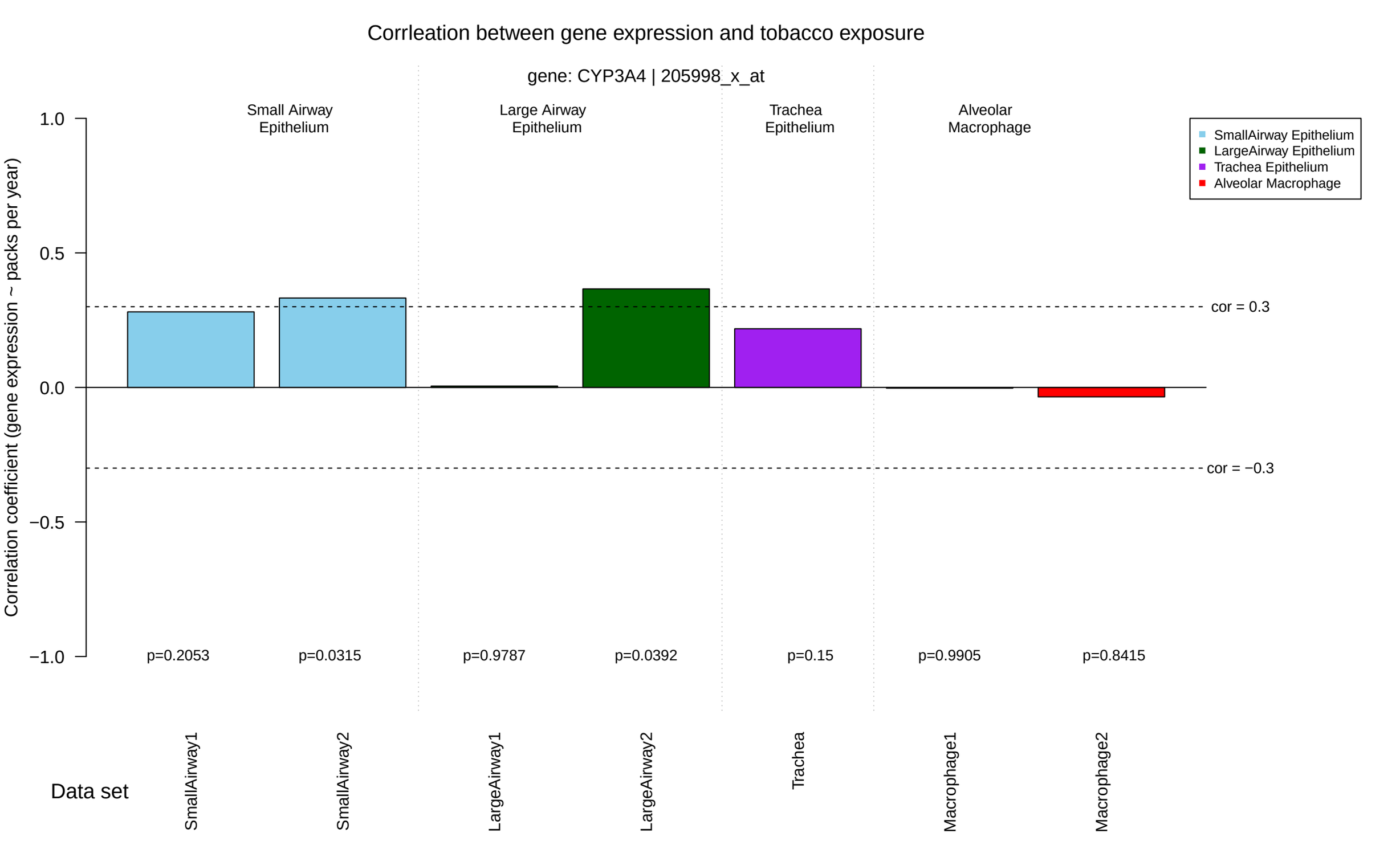

set: LargeAirway1 (p=0.1563)
CYP3A4 | 205998_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1547)
CYP3A4 | 205998_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0128)
CYP3A4 | 205998_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.775)
CYP3A4 | 205998_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4697)
CYP3A4 | 205998_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP3A4 | 205998_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2053
p=0.0315
p=0.9787
p=0.0392
p=0.15
p=0.9905
p=0.8415
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2